

FasterCap 3D Input Files

Input File Structure

The input file is a sequential list of materials definitions (conductor 'C' statements and dielectric 'D' statements) and of geometric panel descriptions (triangular patch 'T' and quadrilateral patch 'Q'). Moreover, comments are allowed ('*' elements) as well as naming definitions ('N' elements)

Definitions can be nested in a hierarchical fashion to be able to re-use the geometric descriptions and to ease the construction of the models.

Panel geometric definitions ('Q' and 'T' elements) in the root file are assumed by default to belong to conductors embedded in a dielectric medium with unit permittivity.

All input files begin with a comment line. For compatibility and to ease possible file merging, you are recommended to start the line with a '*' character; anyway this is not mandatory, and the first line of any input file is simply ignored.

Remark: since FasterCap is able to automatically refine the geometry, it is best working with **an input mesh as coarse as possible**. This is a main difference with FastCap2, that needs a carefully pre-refined input mesh in order to produce accurate results.

An input file example is shown here below.

File *cubes.txt*:

```
* Array of two cubes, in air
*
* conductor | dielectric | offset
* file name | constant   | in space
*
C cube.txt   1.000000    0.0 0.0 0.0
C cube.txt   1.000000    2.0 0.0 0.0
```

The above input file references hierarchically the file *cube.txt*. This file contains a geometric description of a unit cube, as shown here below.

File *cube.txt*:

```
* 1mX1mX1m unit cube
*
* conductor      | 3D coordinates of the four corners
* name to which | of the quadrilateral Q patch
* the Q patch   |
* belongs       |
*
Q mycube        1.0 1.0 0.0  1.0 0.0 0.0  1.0 0.0 1.0  1.0 1.0 1.0
Q mycube        0.0 1.0 0.0  1.0 1.0 0.0  1.0 1.0 1.0  0.0 1.0 1.0
Q mycube        1.0 0.0 0.0  0.0 0.0 0.0  0.0 0.0 1.0  1.0 0.0 1.0
Q mycube        0.0 0.0 0.0  0.0 1.0 0.0  0.0 1.0 1.0  0.0 0.0 1.0
Q mycube        0.0 0.0 0.0  1.0 0.0 0.0  1.0 1.0 0.0  0.0 1.0 0.0
Q mycube        0.0 0.0 1.0  1.0 0.0 1.0  1.0 1.0 1.0  0.0 1.0 1.0
```

The following sections explain in detail the meaning of each statement.

